



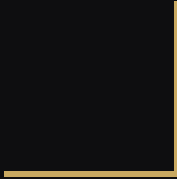
DATABASE-DRIVEN INFORMATION SYSTEM

THE CONTROL GROUP CONSORTIUM

Lab Management System

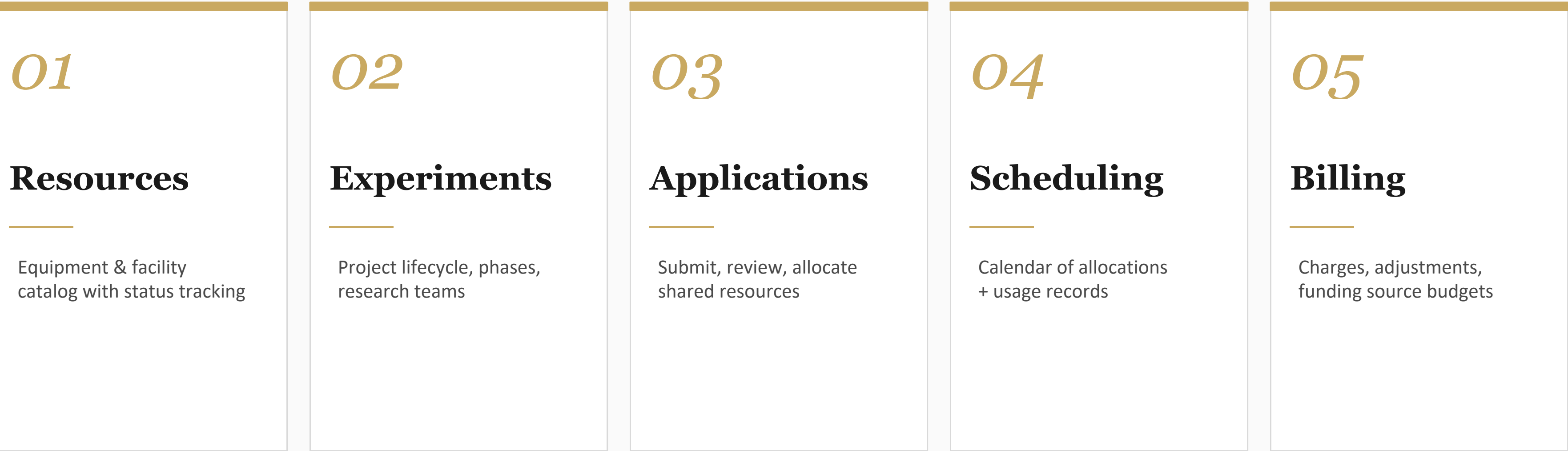
Group 7
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CSCI UA.0060 · Spring 2026



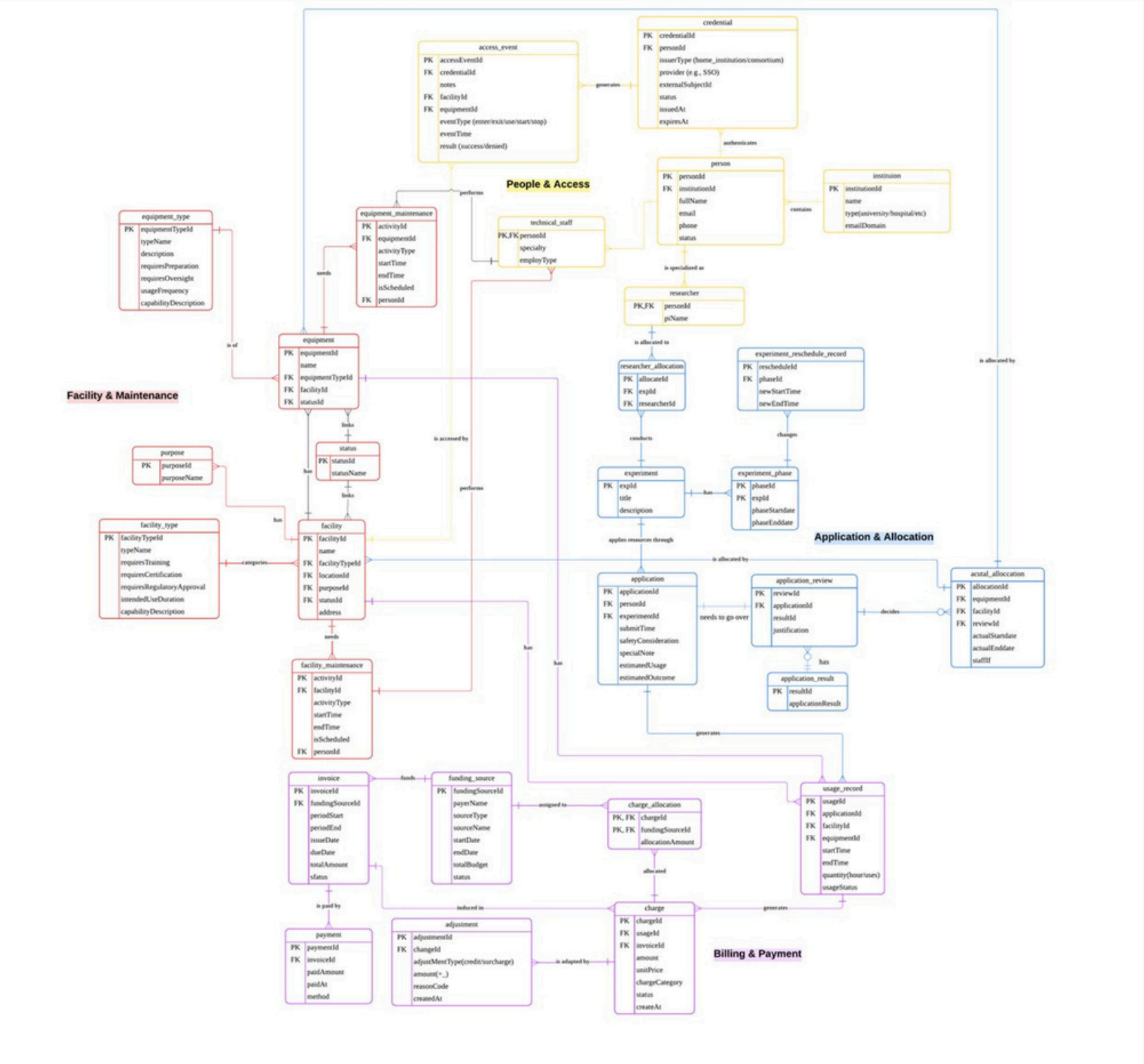
What We Built

A consortium that shares research facilities, equipment, and staff across institutions — our system tracks the full lifecycle from request to billing.



26 tables · 3 triggers · ~1 quarter of activity tracked

Entity Relationship Diagram



4 LOGICAL SECTIONS

● People & Access

person · institution · credential · occupation · access_event

● Application & Allocation

experiment · application · review · actual_allocation · phase

● Facility & Maintenance

facility · equipment · maintenance · types · purpose

● Billing & Payment

charge · adjustment · payment · invoice · funding_source

26 TABLES · 3 TRIGGERS

[View interactive ERD on Lucidchart](#)

From ERD to Database

1

ERD → schema.sql

26 CREATE TABLE

2

Base data (insert.sql)

Lookup + reference rows

3

Mock data populated

1 quarter of activity

```
167 CREATE TABLE application (  
168     applicationId      INT PRIMARY KEY AUTO_INCREMENT,  
169     personId           INT NOT NULL,  
170     experimentId       INT NOT NULL,  
171     submitTime         DATETIME NOT NULL,  
172     safetyConsideration TEXT,  
173     specialNote        TEXT,  
174     estimatedUsage      VARCHAR(100),  
175     estimatedOutcome    TEXT,  
176     equipmentId        INT NOT NULL,  
177     facilityId         INT NOT NULL,  
178     staffId            INT NOT NULL,  
179     FOREIGN KEY (personId) REFERENCES person(personId),  
180     FOREIGN KEY (experimentId) REFERENCES experiment(experimentId),  
181     FOREIGN KEY (equipmentId) REFERENCES equipment(equipmentId),  
182     FOREIGN KEY (facilityId) REFERENCES facility(facilityId),  
183     FOREIGN KEY (staffId) REFERENCES person(personId)  
184 );
```

application table — schema.sql · enforces FK to person, experiment, equipment, facility

MOCK DATA POPULATED

- 20

equipment items
- 12

experiments
- 19

applications submitted
- 15

billable charges

3 TRIGGERS guard data integrity → actual_allocation only allowed when application is approved

Populating Base Data

1

ERD → schema.sql

26 CREATE TABLE

2

Base data (insert.sql)

Lookup + reference rows

3

Mock data populated

1 quarter of activity

```
46 -- institution (10 rows)
47 INSERT INTO institution (institutionId, name, type, emailDomain) VALUES
48 (1, 'MIT', 'university', 'mit.edu'),
49 (2, 'Stanford University', 'university', 'stanford.edu'),
50 (3, 'Johns Hopkins Hospital', 'hospital', 'jhmi.edu'),
51 (4, 'Caltech', 'university', 'caltech.edu'),
52 (5, 'Mayo Clinic', 'hospital', 'mayo.edu'),
53 (6, 'ETH Zurich', 'university', 'ethz.ch'),
54 (7, 'National Institutes of Health', 'research_institute', 'nih.gov'),
55 (8, 'University of Tokyo', 'university', 'u-tokyo.ac.jp'),
56 (9, 'Max Planck Institute', 'research_institute', 'mpg.de'),
57 (10, 'Cambridge University', 'university', 'cam.ac.uk');
58
```

institution table — base reference data for 10 participating organizations

WHAT BASE DATA INCLUDES

- Lookup tables

status, purpose, application_result, charge_category
- Reference types

facility_type, equipment_type, occupation
- Institutions

10 universities, hospitals & research centers
- Funding sources

10 grants & departmental budgets

Stable foundation → enables realistic mock data in step 3

Key Design Decisions

From our 9-decision log — three that shaped how the system behaves.

Equipment OR Facility

THE PROBLEM

Each **application** targets one resource — but originally both FK columns were NOT NULL.

OUR SOLUTION

Made both nullable; added CHECK to enforce exactly one is filled.

Reallocate preserves history

THE PROBLEM

Moving **equipment** to a new facility shouldn't erase where it was before.

OUR SOLUTION

Set end_date on current row, INSERT new assignment row. Full audit trail.

Track funding utilization

THE PROBLEM

Managers need to see how much each funding pool has consumed at a glance.

OUR SOLUTION

Each funding source shows a live progress bar: used / budget / period — surfaced from charge_allocation aggregations.

Turning Records into Insight

12+ queries across two purposes — daily operations and consortium-wide analytics.

OPERATIONAL

Day-to-day workflow

- Equipment list — filter by status / type
- Equipment detail with usage + maintenance history
- Facility detail with currently-assigned equipment
- Experiment phases, team members, linked applications
- Application detail — review + actual allocation
- Allocation calendar (who occupies what, when)
- Usage records with credential issuer info
- Charge detail — adjustments + payment + funding split

ANALYTICAL

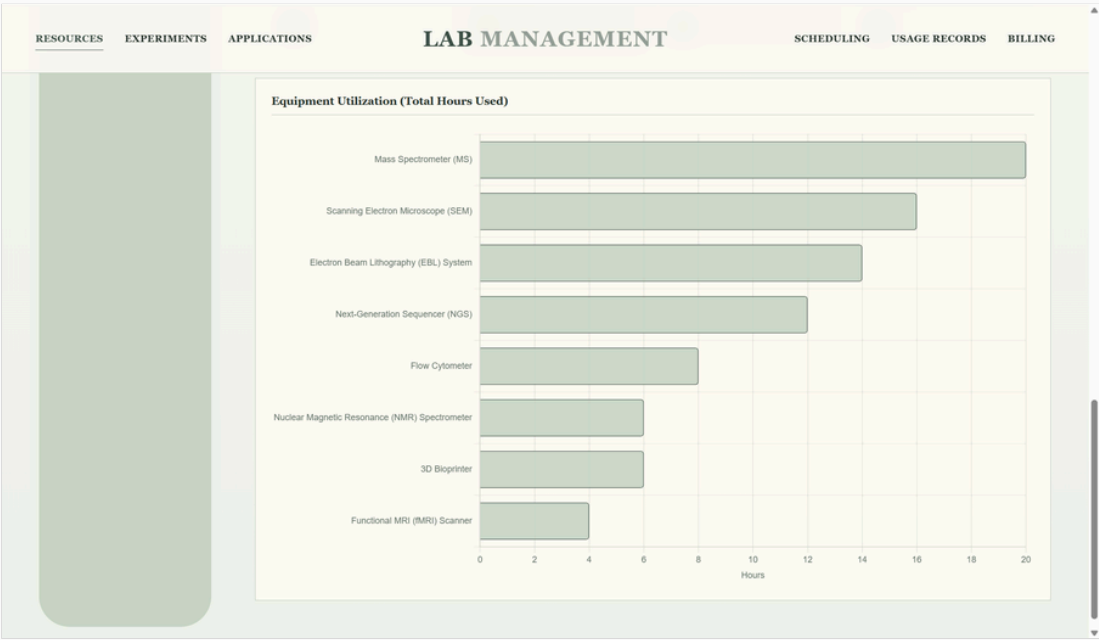
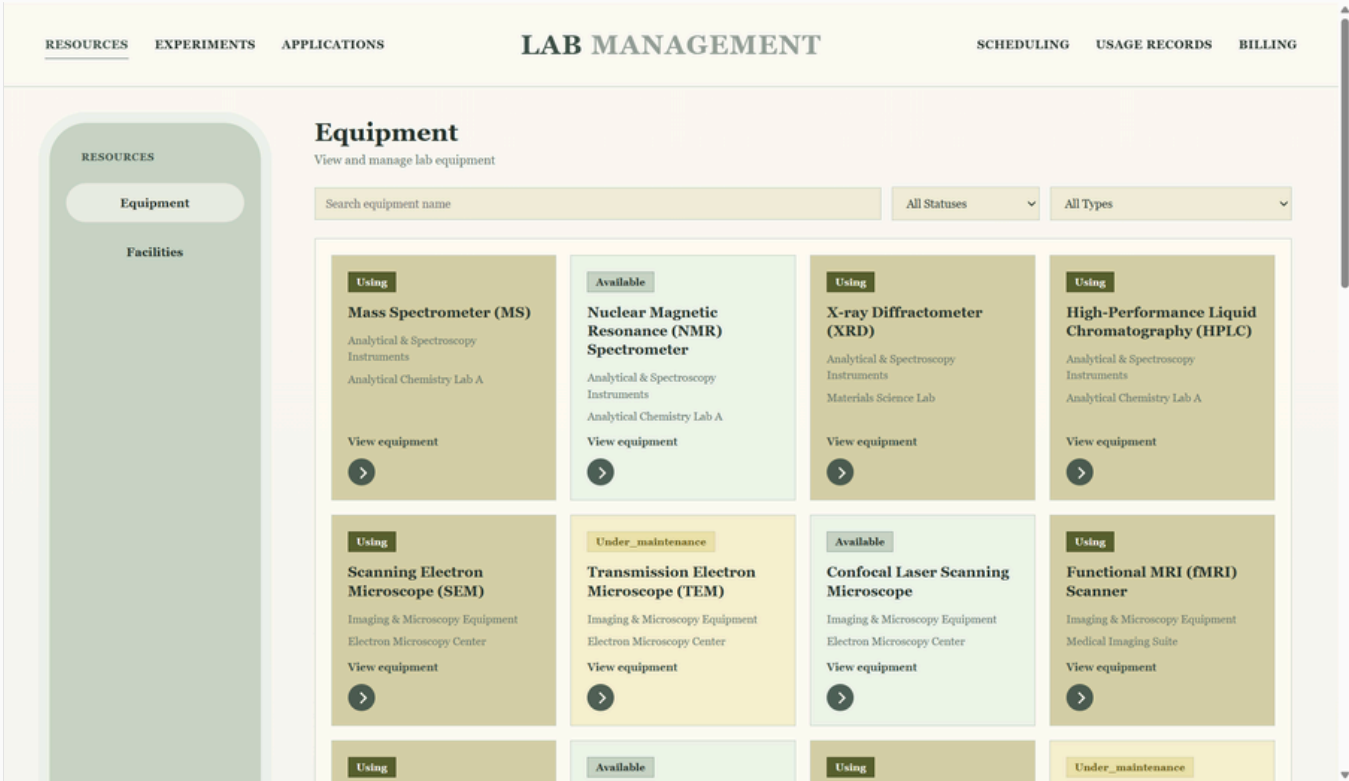
Aggregations + trends

- Application status distribution → donut chart
- Monthly application trend → line chart
- Equipment status breakdown (4 status buckets)
- Equipment utilization — top 8 by usage hours
- Funding source budget utilization (% used)
- Maintenance workload per facility / equipment

Resources: Equipment & Facility

WHAT YOU'LL SEE

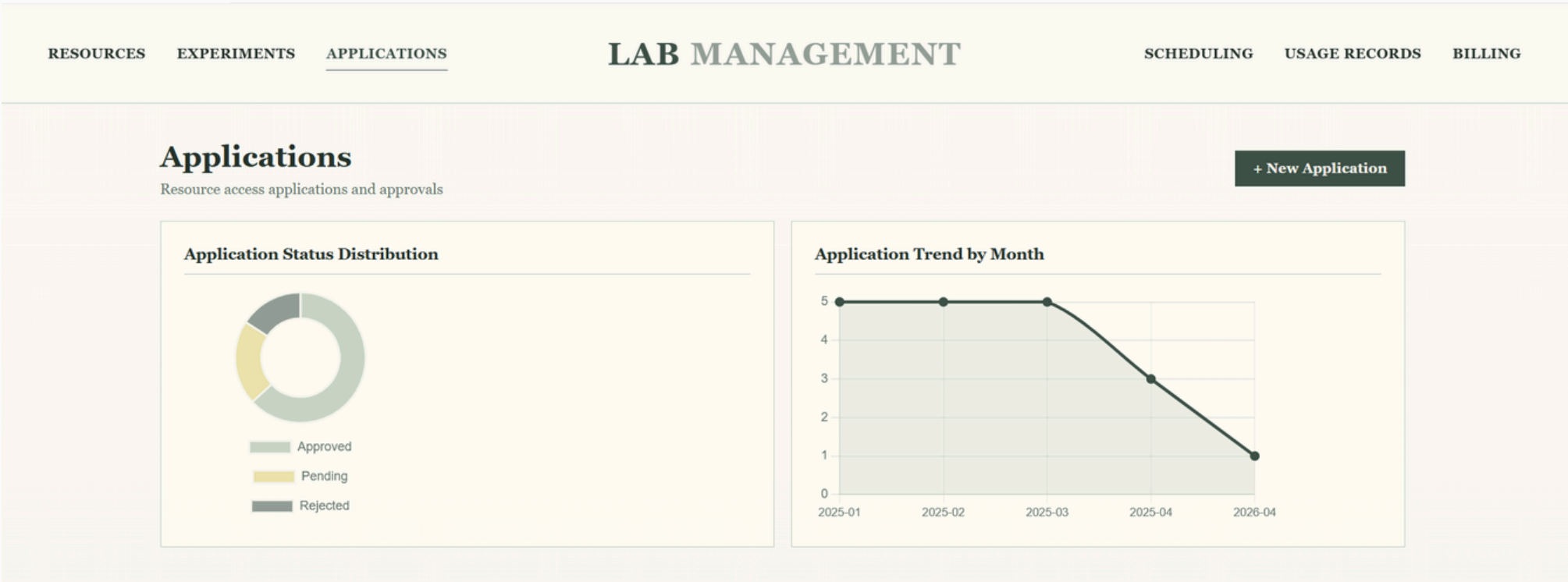
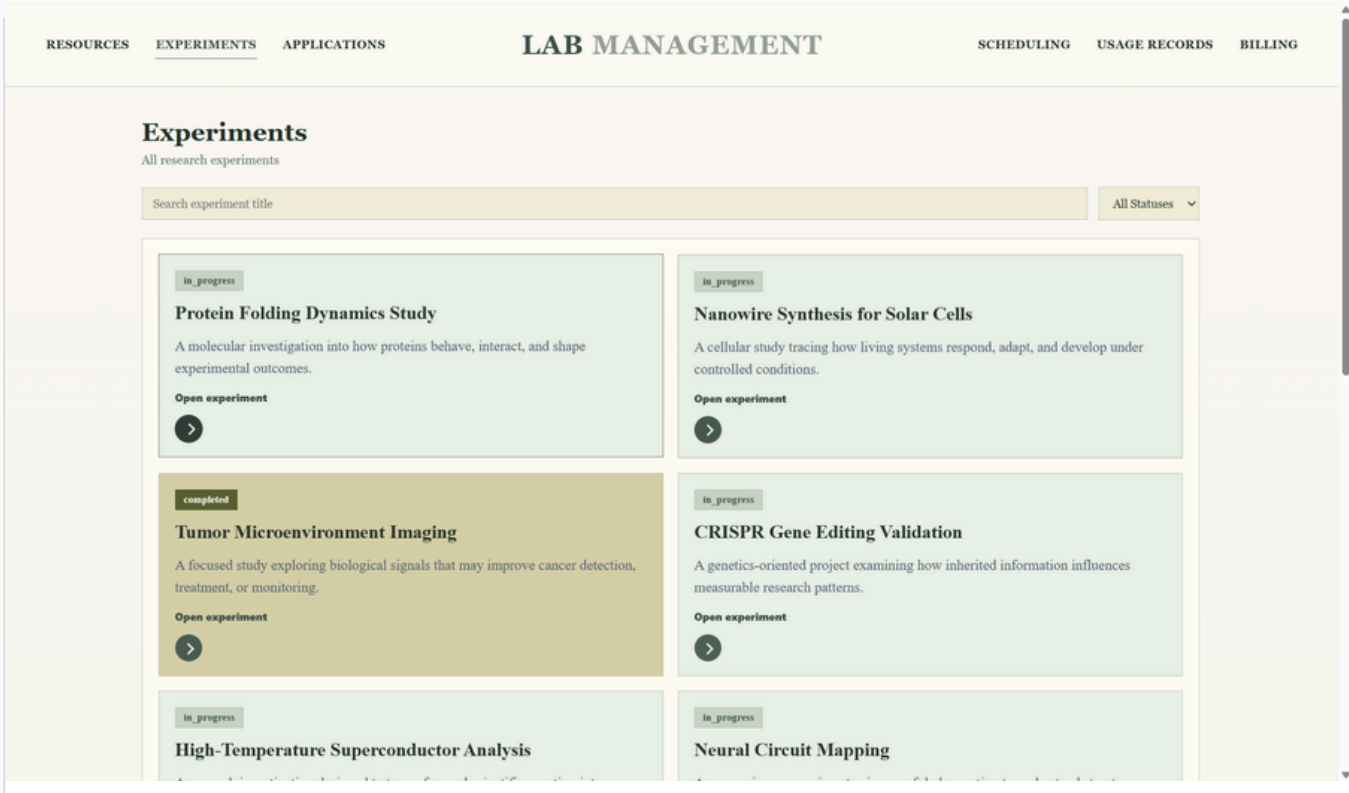
- Equipment list with live status
- Filter by status / type
- 4 stat cards — Available / In Use / Maintenance / Total
- Reallocate modal — preserves history
- Utilization chart — top 8 by hours



Experiments & Applications

WHAT YOU'LL SEE

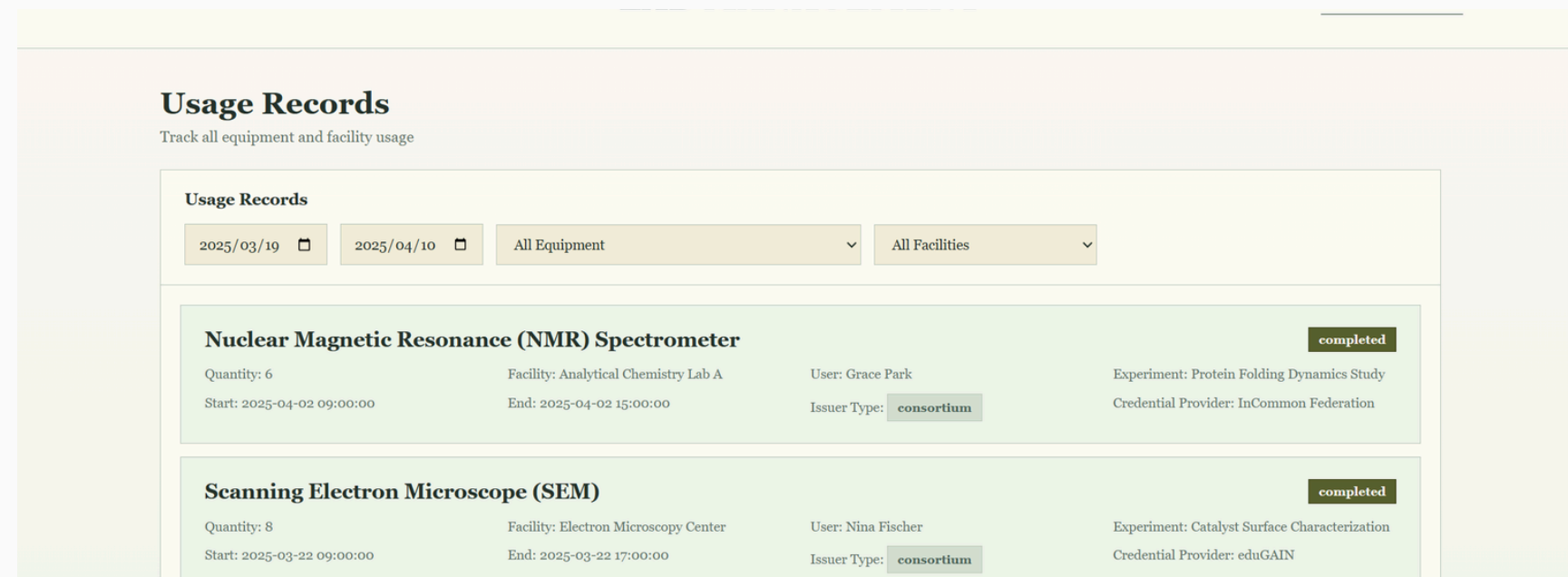
- Experiment phase timeline
- Team members + home institution
- Linked applications (clickable)
- Status donut + monthly trend
- Make Application — equipment / facility toggle



Scheduling: Allocations & Usage

WHAT YOU'LL SEE

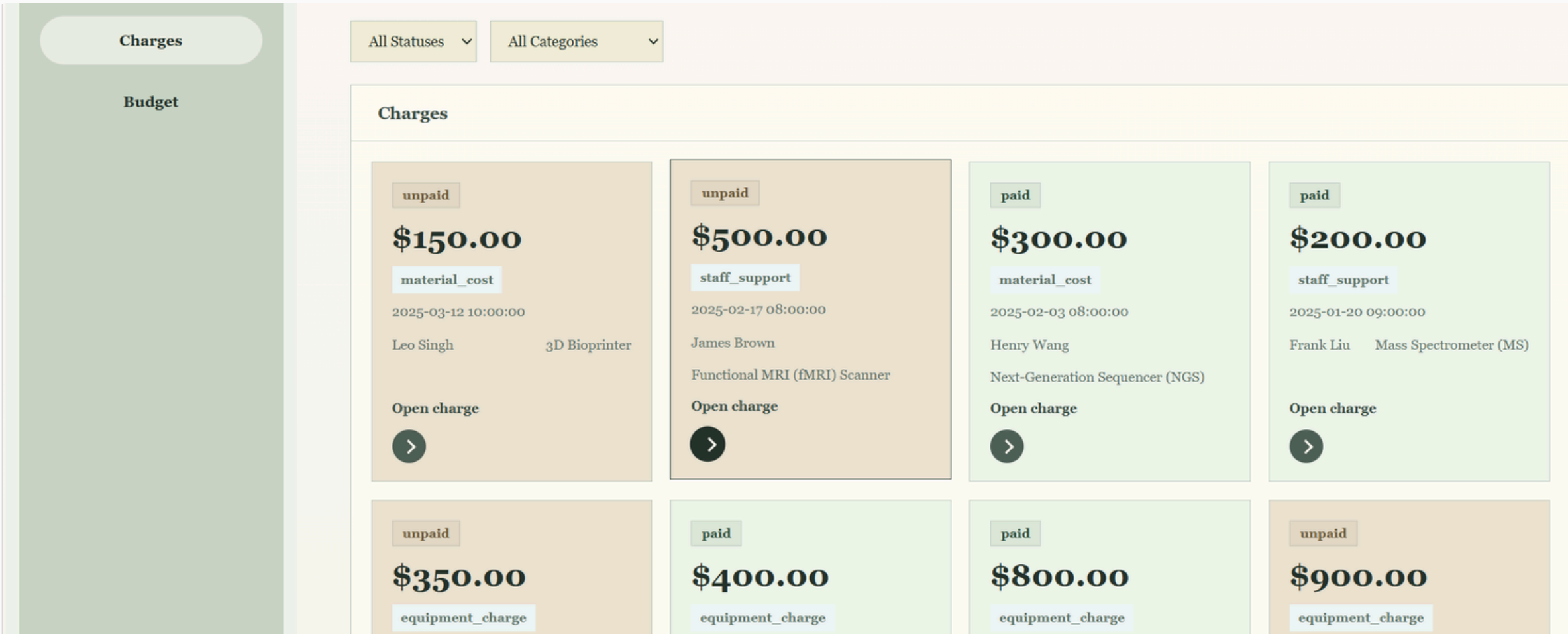
- Calendar view — color-coded lines
- Click a date → see allocations
- Usage records with credential issuer
- home_institution vs consortium
- Cross-references facility + equipment



Billing: Charges & Budget

WHAT YOU'LL SEE

- Charge list — paid / unpaid status
- Categories: equipment / staff / material
- Linked to person, equipment, usage date
- Funding source progress bar
- Used vs budget vs active period

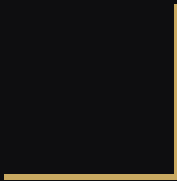




L I V E

Demonstration

Flask + SQLite



CONCLUSION

Thank You

26

tables modeled

3

data-protection triggers

12+

analytical queries

5

core modules shipped

THE TEAM

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